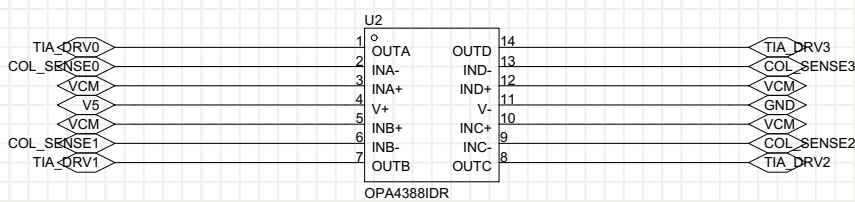
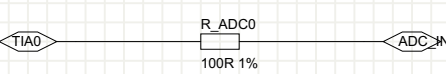
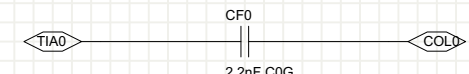
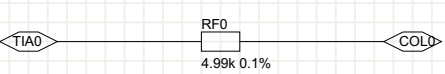
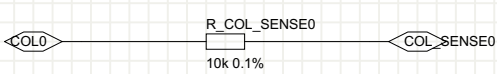
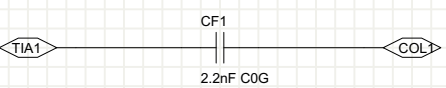
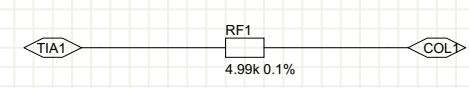
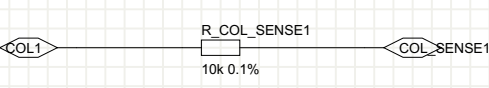
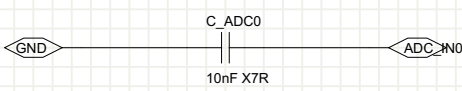
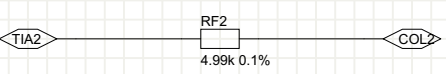
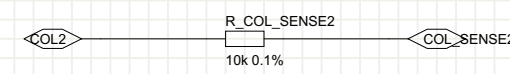
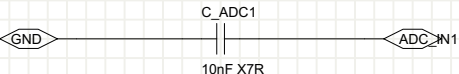
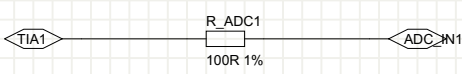
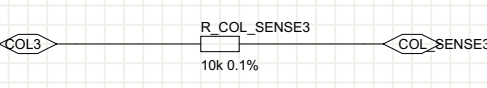
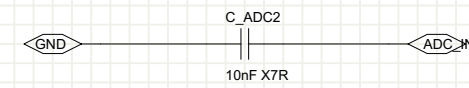
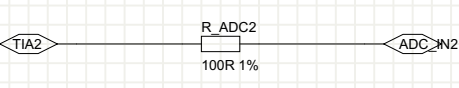
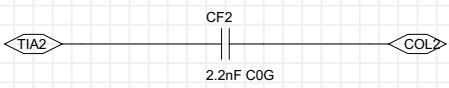
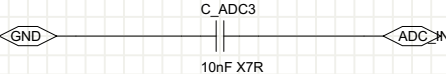
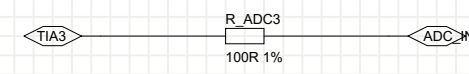
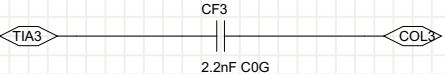
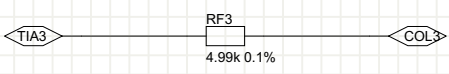
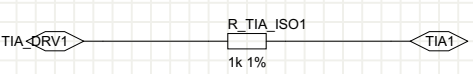
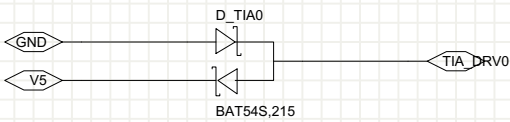
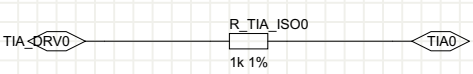
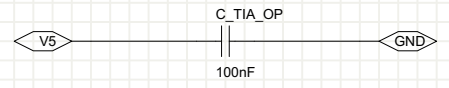
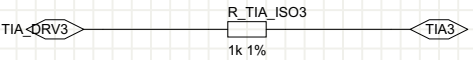
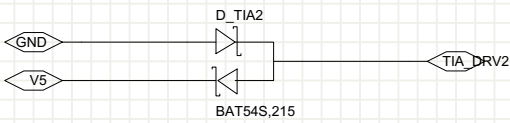
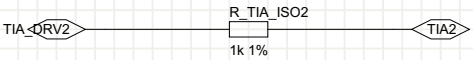
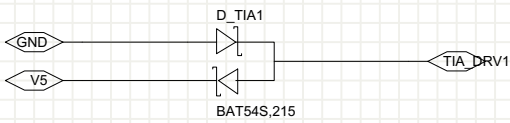
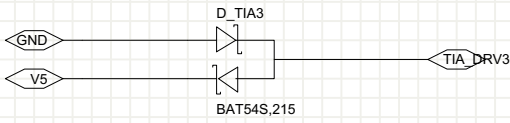
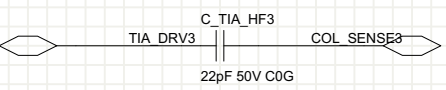
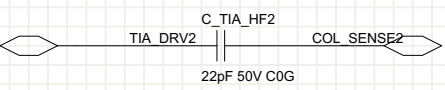
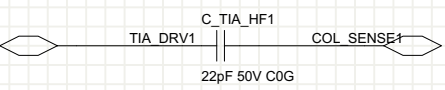
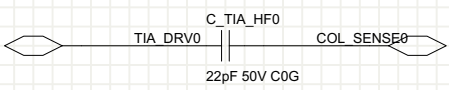


R2 REVIEW ONLY - Power_Reference - 1/6

R2 REVIEW ONLY - Rows_8lead_Interface - 2/6

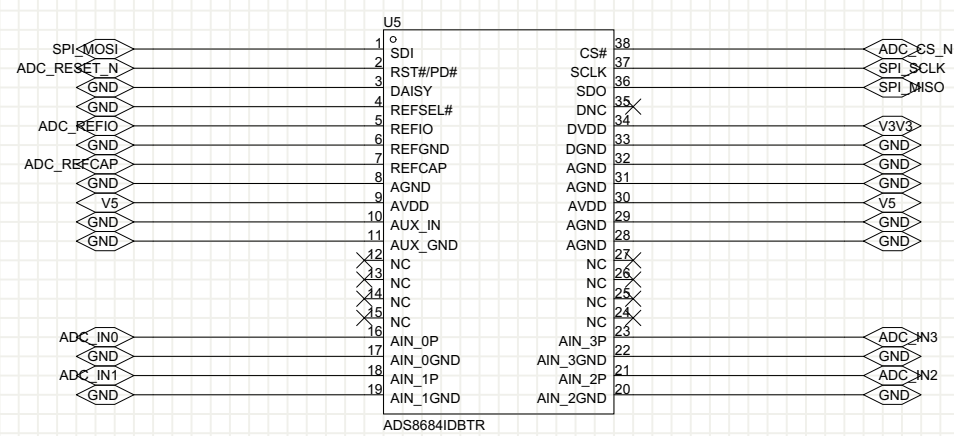
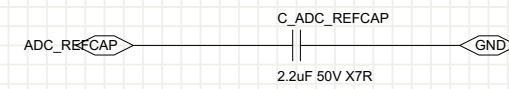
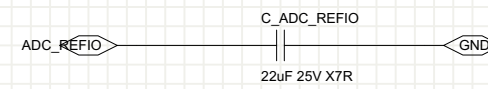
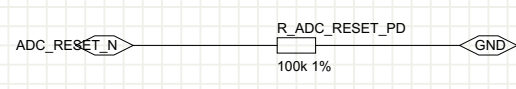
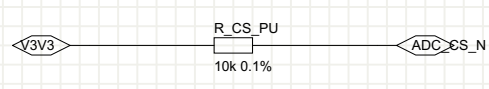
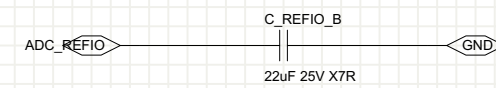
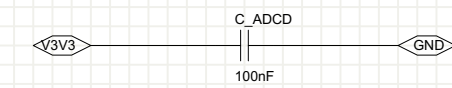
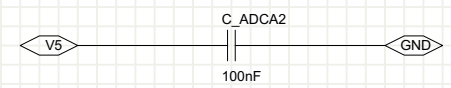
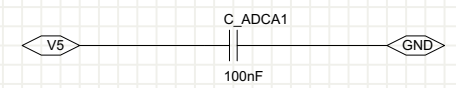
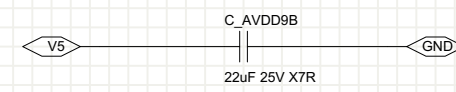
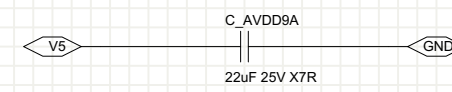
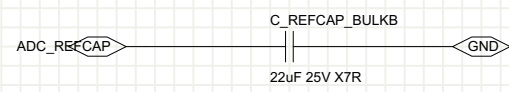
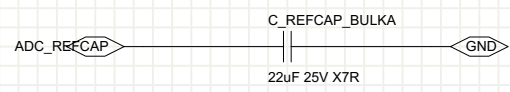
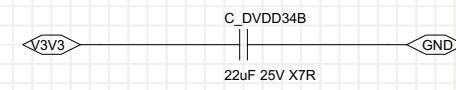
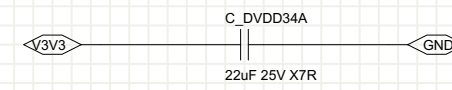
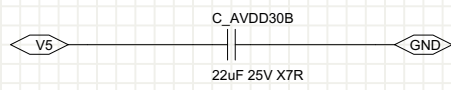
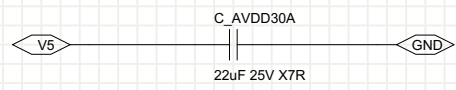
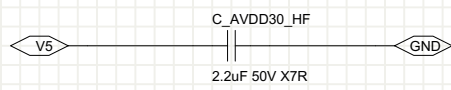
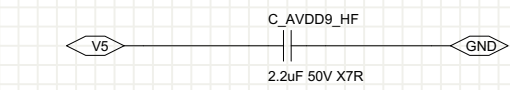


COL -> 10k -> OPA minus; internal input-clamp current bounded, fault/thermal and macro loop HOLD

OPA output -> 1k -> TIA tap; all ADC/filter/feedback takeoffs after isolation

Four continuous TIAs; Rf4.99k || Cf2.2nF from isolated TIA tap to COL

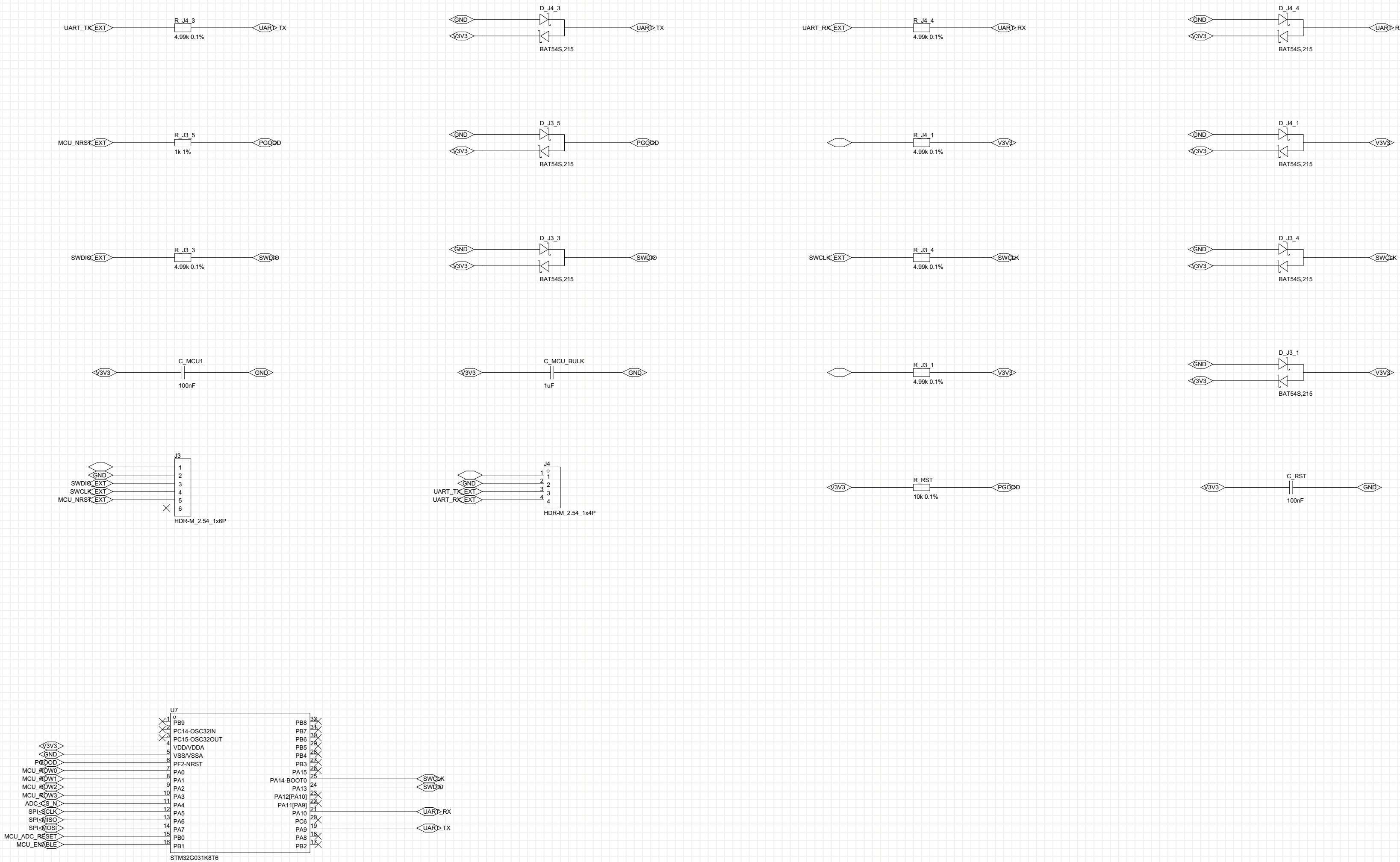
R2 REVIEW ONLY - Continuous_Column_TIAs - 3/6



Effective capacitance and placement must meet datasheet: REFERENCE_CAPACITANCE_HOLD

AVDD pins9/30 each44uF+2.2uF nominal; DVDD pin34 44uF nominal

Internal4.096V ADC reference; REFIO and REFCAP independently bulk bypassed



Offline SPI: 0-5.12V range0x06, tagged48clk frames, 40kSPS / hardware timing HOLD

External SWD/UART/sense lines all4.99k current limited, clamps and rail sink

GPIO PB1 pin16 = MCU_ENABLE; PGOOD resets MCU; Schmitt reshapes hardware gate edge

R2 REVIEW ONLY - MCU_Protected_Interfaces - 5/6

